

BTEC Level 3 Applied Science • Unit 1 Chemistry • Answer sheet

Electron Configuration — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.

A–E

1. 2, 8 and 8.
2. s: 2. p: 6.
3. Group 1: one outer electron. Group 7: seven outer electrons.
4. N 2,5; Mg 2,8,2; Cl 2,8,7; K 2,8,8,1.
5. O $1s^2 2s^2 2p^4$; Al $1s^2 2s^2 2p^6 3s^2 3p^1$; Ca $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$.
6. Hund: one electron in each equal-energy orbital before pairing. N $2p^3$ three single arrows; O $2p^4$ one pair plus two singles.

Orbital box diagrams — boxes and spin arrows drawn

Each box is one orbital. An up arrow and a down arrow are electrons of opposite spin.

Helium $1s^2$ 

1s

Pauli: two electrons in one orbital must have opposite spin.

Nitrogen $2p^3$ 

2px

2py

2pz

Hund: one electron in each equal-energy orbital before pairing.

Oxygen $2p^4$ 

2px

2py

2pz

The fourth 2p electron pairs in one box. The other two stay unpaired.

7. O^{2-} 2,8 / $1s^2 2s^2 2p^6$; Al^{3+} 2,8 / $1s^2 2s^2 2p^6$; Cl^- 2,8,8 / $1s^2 2s^2 2p^6 3s^2 3p^6$.

8. Calcium is 2,8,8,2. After 2,8,8 the next two electrons occupy the fourth shell / 4s, not a 10-electron third shell.

9. Mg is 2,8,2. Losing two electrons leaves 2,8, a full outer shell.

10. Mg^{2+} is 2,8 or $1s^2 2s^2 2p^6$. Full outer-shell / noble-gas arrangement is stable. [3]

11. Oxygen is 2,6. Gaining two electrons gives 2,8. [3]
12. Na s-block; F p-block; Fe d-block, with one justification. [3]
13. P is $[\text{Ne}] 3s^2 3p^3$. P^{3-} is $[\text{Ne}] 3s^2 3p^6$. The ion has gained three electrons to fill 3p.